

RAK12032 WisBlock 3-Axis Accelerometer Sensor Datasheet

Overview

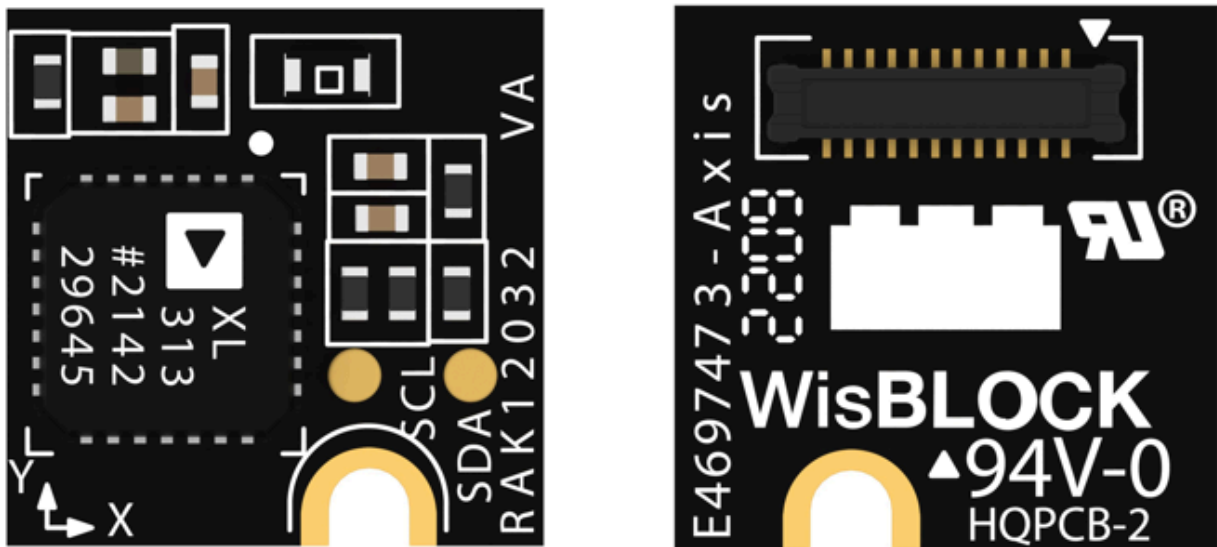


Figure 1: RAK12032 WisBlock 3-Axis Sensor

Description

RAK12032 is a WisBlock 3-axis Accelerometer Sensor Module that extends the WisBlock system with an ADXL313 3-axis accelerometer from Analog Devices. It comes with a ready-to-use software library and tutorial, making it easy to build a motion-sensing data acquisition system. The motion-sensing data is interfaced via I2C. Additionally, RAK12032 can be mounted to the sensor slot of the WisBlock Base board.

Product Features

- Sensor specifications
 - Based on ADXL313

- Measurement range: $\pm 0.5g$, $\pm 1g$, $\pm 2g$, $\pm 4g$
 - High resolution (up to 13-bit)
 - Ultra-low power (scales automatically with data rate), as low as 30 μA in measurement mode
 - Low noise performance: 150 $\mu g/Hz$ typical for X-Y axes
 - Built-in motion detection functions for activity/inactivity monitoring
 - I2C digital interfaces
 - 3.3 V power supply
 - Operating temperature range: $-40^{\circ}C$ to $+85^{\circ}C$
- **Size**
 - 10×10 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK12032 module on a [WisBlock Base](#) board. The RAK12032 module can be mounted on the slots: C, D, E, & F.

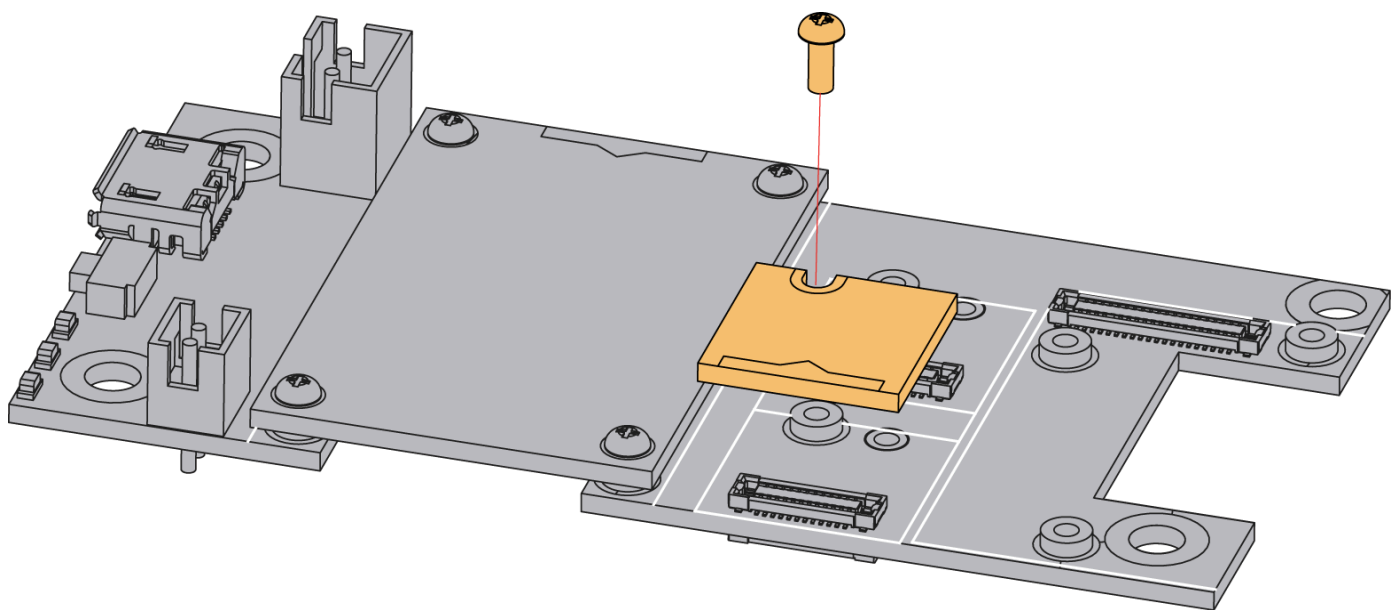


Figure 2: RAK12032 WisBlock 3-Axis Sensor Mounting

NOTE

When the RAK12032 is used with other sensor modules powered by **3V3_S** together, it can not be mounted on slot **A** and slot **B**.

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK12032 WisBlock 3-Axis Sensor.

Chipset

Vendor	Part Number
Analog Devices	ADXL313

Pin Definition

The RAK12032 WisBlock 3-Axis Sensor comprises a standard WisBlock connector. The WisBlock connector allows the RAK12032 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 3**.

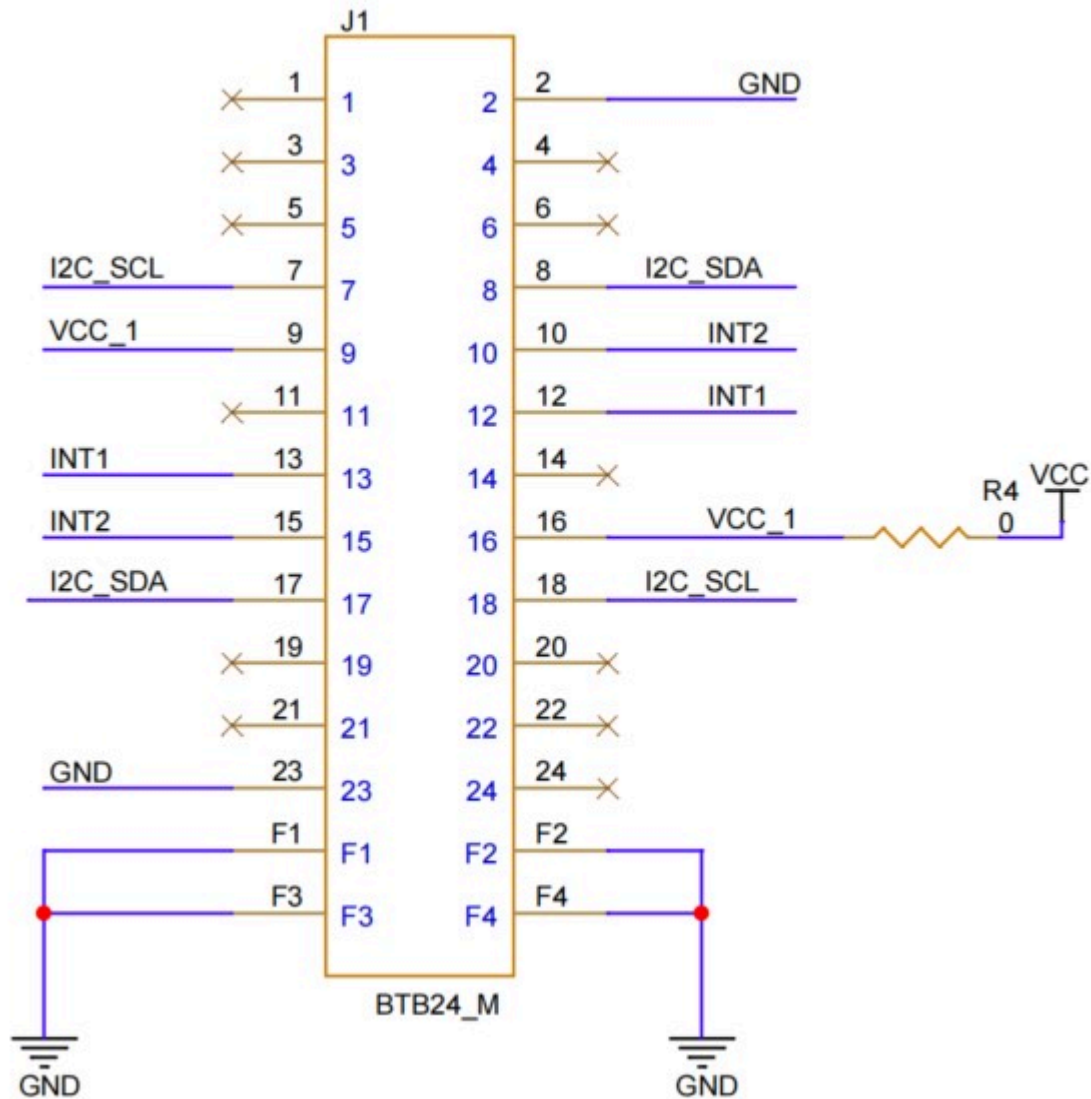


Figure 3: RAK12032 WisBlock 3-Axis Sensor Pinout Diagram

NOTE

The following are connected to WisBlock connector:

- I2C-related pins
- INT1
- INT2
- VDD
- GND

INT2 pin at **Pin 10** and INT1 pin at **Pin 12** will depend on where sensor slot the module is plugged in. The table shows the compatible pins used by different sensor slots:

INT2 (Interrupt 2 Output Pin)

Slot C	Slot D	Slot E	Slot F
IO4	IO6	IO3	IO5

INT1 (Interrupt 1 Output Pin)

Slot C	Slot D	Slot E	Slot F
IO3	IO5	IO4	IO6

Electrical Characteristics

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VCC	Power supply voltage	-	3.3	-	V
IDD	Measure current	30	-	300	uA
IDDS	Standby current	-	76.2	-	nA

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK12032 module.

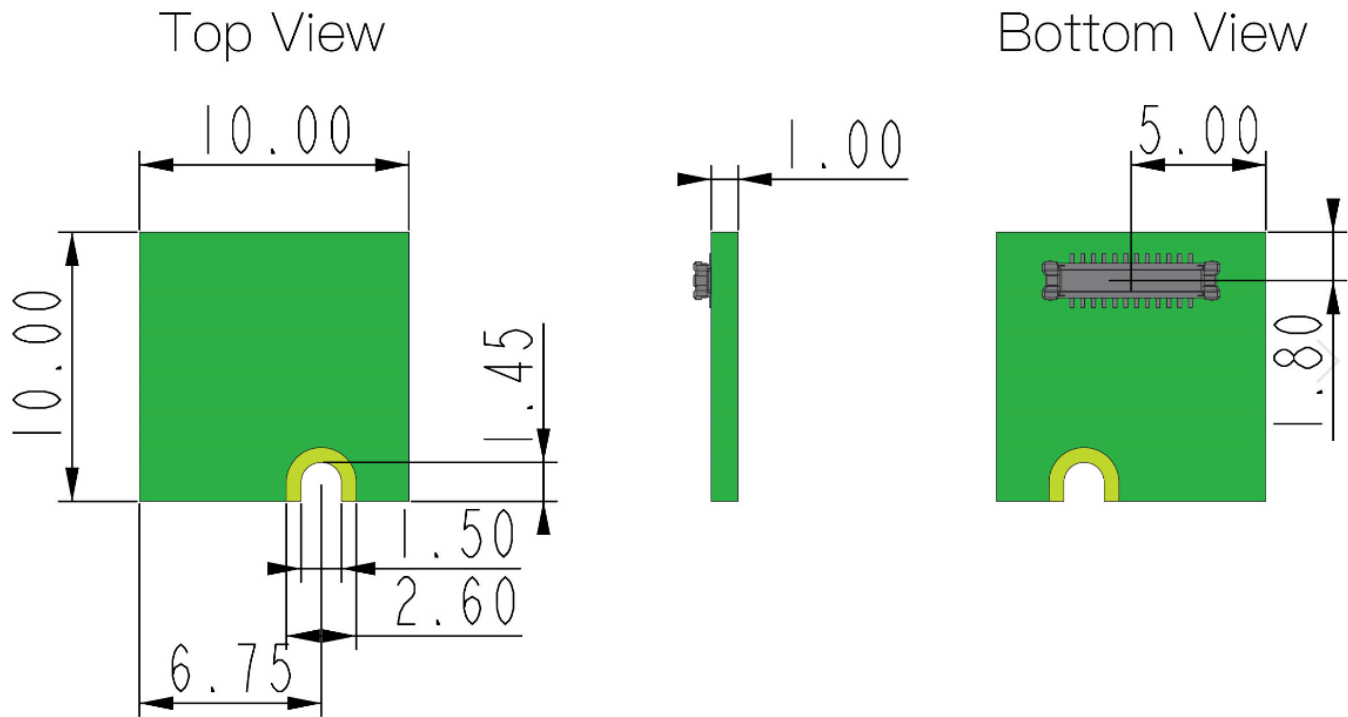


Figure 4: RAK12032 WisBlock 3-Axis Sensor Mechanic Drawing

WisConnector PCB Layout

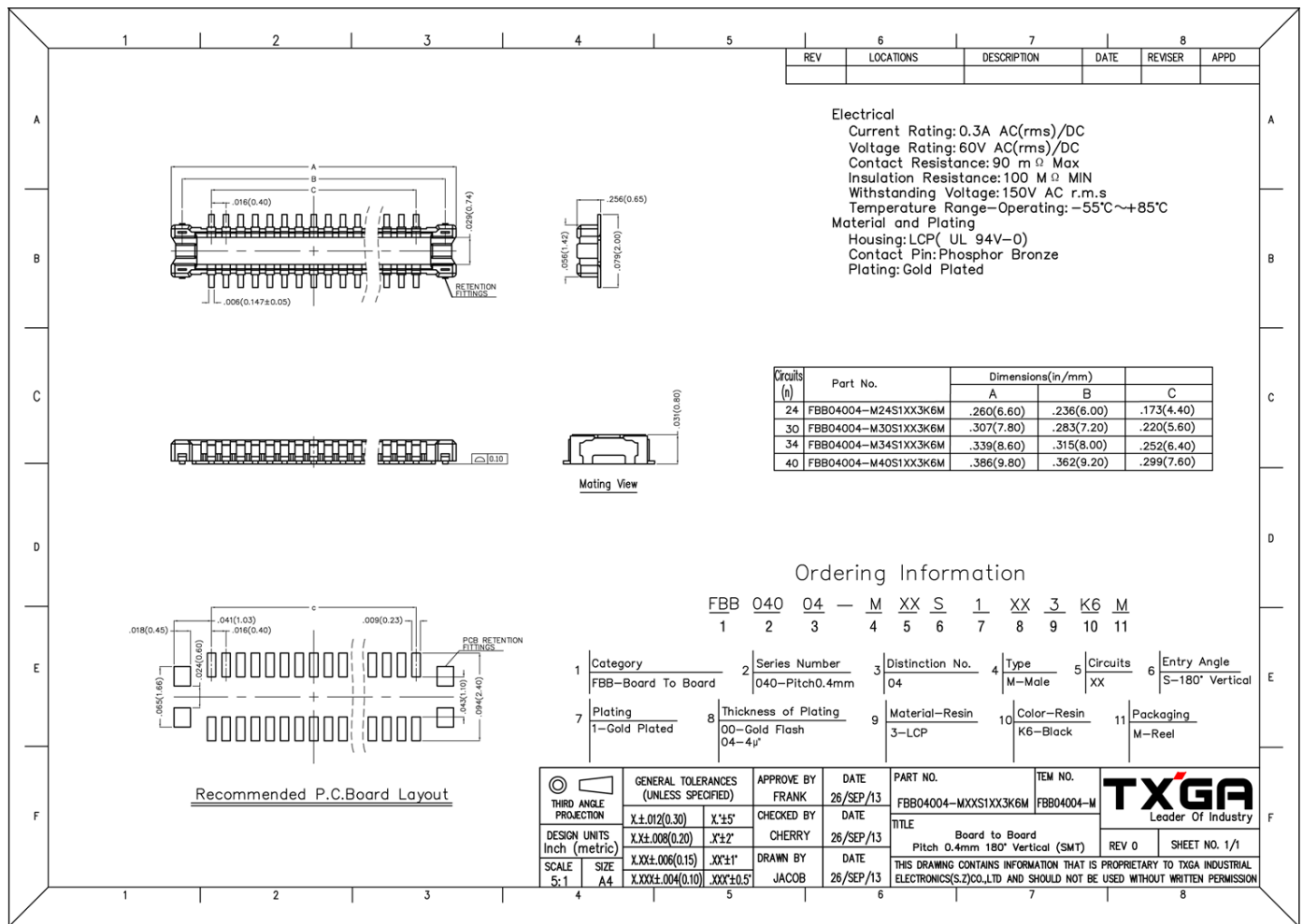


Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK12032 module.

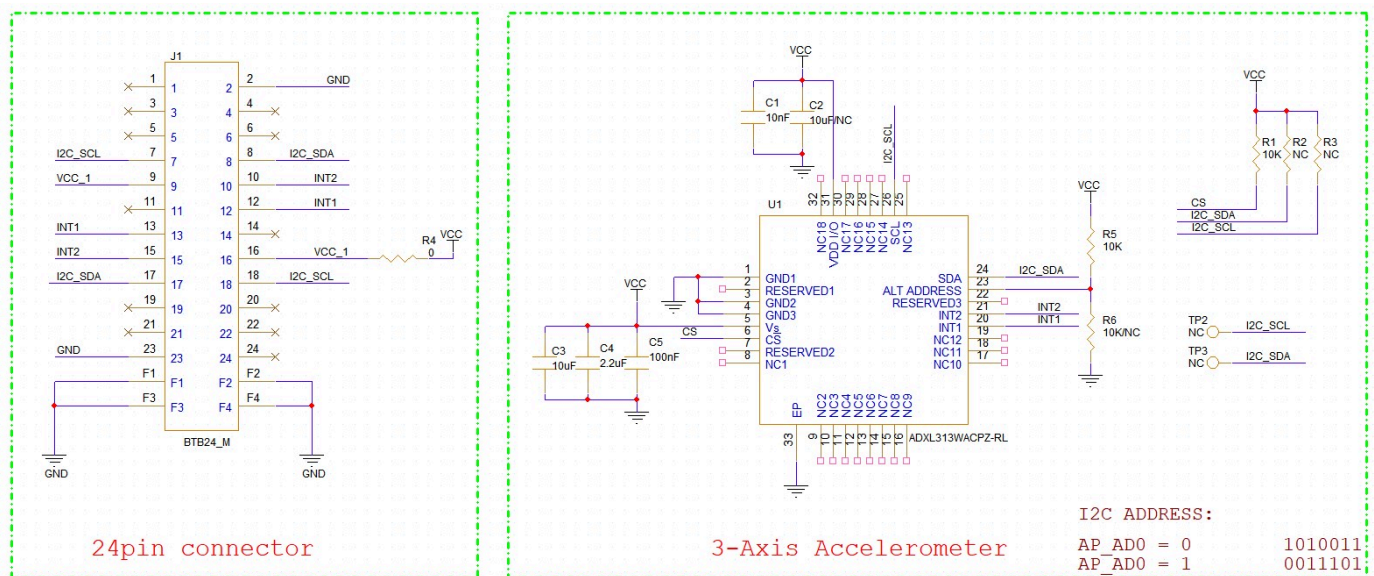


Figure 6: RAK12032 WisBlock 3-Axis Sensor schematics